Department of Physics

PCS 125–Waves and Fields

Please insert in the circle the initial of your last name inside

Please Print Clearly:

Student:_____ _____

Last Name ***First Name***

Student Number: _____ **Student Section:** _____

Final exam – April 22,2015

Check the blank box in from of the name of your professor:

Dr. Chithrani		Dr. Douplick		Dr. Manzhos		Dr. Hu		Dr. Toronov

Read the following attentively

- Duration of the exam: 3hours (180 minutes).
- The exam has 13 pages (including the front page and the formula sheet page).
- The exam consists of 21multiple choice questions, each worth one mark. The maximum exam mark is 21.
- Note that for each multiple choice question:
  - You have to provide, in this booklet, a justification for your answer.
  - The work shown in your exam booklet may be checked. If no work or correct argument is shown to validate your answer, the resulting mark may be zero, even if the answer in the scantron is correct.
  - Transfer your choices to the scantron only when you're sure of them. Some answers may be rounded off.
- All Ryerson exam rules apply.
- Sharing of pens, erasers and calculators is not permitted.
- Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.
- A formula sheet is provided with the test booklet. No other sheets are allowed.
- The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) it will be confiscated for the duration of the test

_____

Student signature (By signing here you indicate that you read all the instructions. No signature, no mark!)

1. To double the total energy of a mass oscillating at the end of a spring with amplitude A, we need to

A) increase the angular frequency by $\sqrt{2}$
B) increase the amplitude by $\sqrt{2}$.
C) increase the amplitude by 2.
D) increase the angular frequency by 2.
E) increase the amplitude by 4 and decrease the angular frequency by $\frac{1}{\sqrt{2}}$

2. Consider the following statements

(i). The work done by external agent to separate masses m_1 and m_2, which are at distance r, to infinite distance is

$$G\frac{m_1 m_2}{r}.$$

(ii).The work done by external agent to separate masses m_1 and m_2, which are at distance r, to infinite distance is

$$-G\frac{m_1 m_2}{r}.$$

(iii).The work done by gravitational field to separate masses m_1 and m_2, which are at distance r, to infinite distance is

$$G\frac{m_1 m_2}{r}.$$

(iv).The work done by gravitational field to separate masses m_1 and m_2, which are at distance r, to infinite distance is

$$-G\frac{m_1 m_2}{r}.$$

Select the correct option below:

a. Only (i) is true.
b. Only (iv) is true.
c. Only (i) and (iv) are true
d. Only (iii) is true
e. Only (ii) and (iii) are true.

3. Point A of coordinates (2, 3) m and point B of coordinates (5, 7) m are in a region where the electric field is uniform and given by
E = (4**i** + 3**j**) N/C. What is the potential difference V_A-V_B?

a. 33V
b. 27V
c. -24V
d. +24V
e. None of the above

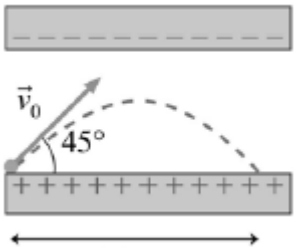
4. A charged particle (q = –8.0 mC), which moves in a region where the only force acting on the particle is an electric force, is released from rest at point A. At point B the kinetic energy of the particle is equal to 4.8 J. What is the electric potential difference V_B-V_A?

a. -0.60 kV
b. +0.60 kV
c. -0.80 kV
d. +0.80 kV
e. -0.48 kV

5. A particle (mass = 6.7×10^{-27} kg, charge = 3.2×10^{-19} C) moves along the positive *x* axis with a speed of 4.8×10^{5} m/s. It enters a region of uniform electric field parallel to its motion and comes to rest after moving 2.0 m into the field. What is the magnitude of the electric field?

a. 2.0 kN/C
b. 1.5 kN/
c. 3.5 kN/C
d. 1.2 kN/C
e. 2.4 kN/C

6. An electron is projected at an angle of $45°$ with a speed of 6.0×10^{6} m/s into the region of upward uniform electric field between two horizontal plates. The electron lands 5.0 cm away from the point at which it was projected into electric field. The magnitude of uniform electric field is



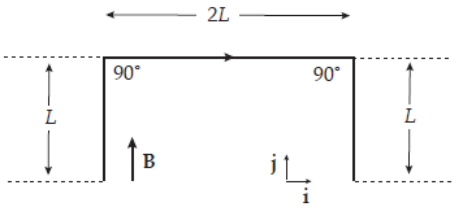
x = 5.0 cm

a. 2.05×10^{3} N/C
b. 4.10×10^{3} N/C
c. 0.410×10^{3} N/C
d. 2.51×10^{5} N/C
e. 2.12×10^{-3} N/C

7. A flute has a length of 58.0 cm. If the speed of sound in air is 343 m/s, what is the fundamental frequency of the flute, assuming it is a tube closed at one end and open at the other?

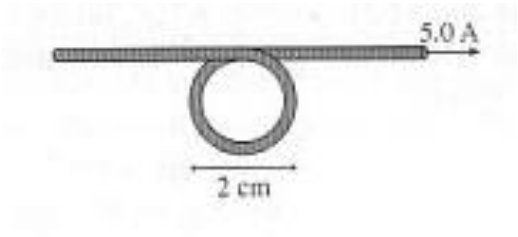
a. 148 Hz
b. 296 Hz
c. 444 Hz
d. 591 Hz
e. none of those answers

8. A straight wire is bent into the shape shown. Determine the net magnetic force on the wire when the current I travels in the direction shown in the magnetic field **B**.



a. $2IBL$ in the $-z$ direction
b. $2IBL$ in the $+z$ direction
c. $4IBL$ in the $+z$ direction
d. $4IBL$ in the $-z$ direction
e. zero

9. A conductor consists of a circular loop and two straight, long sections as shown in figure. The wire lies in the plane of the paper and carries a current of 5.0 A. What is the strength and direction of the magnetic field at the center of the loop?



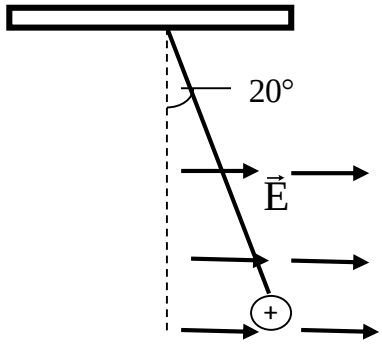
a. 2.1×10^{-4}T (out of the page)
b. 2.1×10^{-4}T (into the page)
c. 4.1×10^{-4}T (out of the page)
d. 4.1×10^{-4}T (into the page)
e. 4.1×10^{-4}T (upward)

10. Two sound waves, from two different sources with the same frequency, 540 Hz, travel in the same direction at 330 m/s. The sources are in phase. What is the phase difference of the waves at a point that is 4.40 m from one source and 4.00 m from the other?

a. 4.12 rad
b. 3.81 rad
c. 1.56 rad
d. -0.52 rad
e. -2.31 rad

11. The figure below shows small charged ball of mass m = 10 g hanging from thread of negligible mass and length 50 cm. A uniform electric field $\vec{E} = 100\ \text{N/C}\ \hat{i}$, indicated by horizontal arrows, causes the thread to make an angle of $20°$ with the vertical. The value of the charge on the ball is

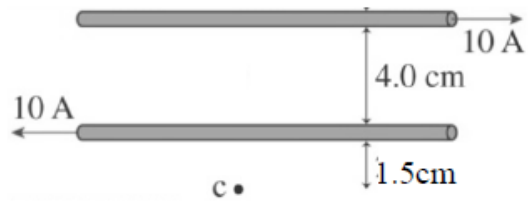
a. 357 μC
b. 269 μC
c. 2693 μC
d. 3567 μC
e. 490 μC



12. A body of mass 5.0 kg is suspended by a spring which stretches 10 cm when the mass is attached. It is then displaced downward an additional 5.0 cm and released. Its position as a function of time is approximately

a. $y = -0.10 \sin 9.9t$
b. $y = 0.10 \cos 9.9t$
c. $y = -0.10 \cos (9.9t + .1)$
d. $y = 0.10 \sin (9.9t + 5)$
e. $y = -0.05 \cos 9.9t$

13. Two wires carry 10A current in the opposite direction. What is the strength and direction of the magnetic field at point c?



(a) 1.7 x10^{-4}T (out of the page) (b)1.7 x10^{-4}T (into the page)
(c) 9.7x10^{-5}T (out of the page)
(d) 9.7x10^{-5}T (into the page)
(e) 9.7x10^{-5}T (downward)

14. A particle (m = 20 mg, q = –5.0 μC) moves in a uniform electric field of 60 N/C in the positive x direction. At t = 0, the particle is moving 30 m/s in the positive x direction and is passing through the origin. The maximum distance beyond x = 0 the particle travels in the positive x direction is

a. 25 m
b. 20 m
c. 15 m
d. 30 m

15. Write the equation of a wave, traveling along the $+x$ axis with an amplitude of 0.02 m, a frequency of 440 Hz, and a speed of 330 m/sec.

a. $y = 0.02 \sin [880\pi(x/330 - t)]$
b. $y = 0.02 \cos [880\pi x/330 - 440t]$
c. $y = 0.02 \sin [880\pi(x/330 + t)]$
d. $y = 0.02 \sin [2\pi(x/330 + 440t)]$
e. $y = 0.02 \cos [2\pi(x/330 + 440t)]$

16. A Hall-effect probe operates with a 140 mA current. When the probe is placed in a uniform magnetic field of magnitude 0.08T, it produces a Hall voltage of 0.6μV. When it is used to measure an unknown magnetic field, the Hall voltage is 0.4 μV. What is the magnitude of the unknown field?

(a) 0.53 mT
(b) 53.3 mT
(c) 33.3 mT
(d) 0.30 mT
(e) 18.8 mT

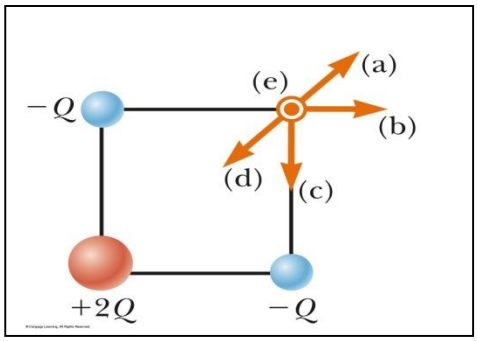
17. Two long, parallel conductors, separated by 10cm, carry currents in the same direction. The first wire carries current I_1 = 5A and the second carries I_2 = 8A.What is the force per unit length exerted by I_1 on I_2?

(a) 10^{-5} N/m (b) $5x10^{-5}$N/m (c) $8x10^{-5}$ N/m (d) $5x10^{-7}$ N/m (e) $8x10^{-7}$N/m

18. An electron travels through a region of space with no acceleration. Which one of the following statements is the best conclusion?

(a) Both **E** and **B** must be zero in that region.
(b) **E** must be zero, but **B** might be non-zero in that region.
(c) **E** and **B** might both be non-zero, but they must be mutually perpendicular.
(d) **B** must be zero, but **E** might be non-zero in that region.
(e) **E** and **B** might both be non-zero, but they must point in opposite directions.

19. Three charged particles are arranged on the corners of a square as shown in the figure below.



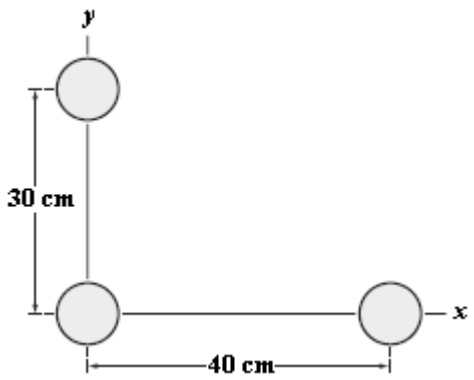
The direction of electric field at the upper right corner

a. is upward and to the right.
b. is straight to the right.
c. is straight downward.
d. is downward and to the left.
e. is perpendicular to the plane of the picture and outward.

20. Two spherical planets of masses *M* and 6M are separated by a distance **d**. The distance (measured from the mass 6M) to a point (located on the straight line which connects the centers of the planets) at which the net gravitational force on a third mass would be zero is

a. 0.29 *d*
b. 0.67 *d*
c. 0.63 *d*
d. 0.71 *d*
e. 0.86 *d*

21. Three 5.0-kg masses are located at points in the *xy* plane as shown in the figure below.



The magnitude of the resultant force (caused by the other two masses) on the mass at $x = -0.40$ m, $y = 0$ is

a. 2.2×10^{-8} N
b. 1.9×10^{-8} N
c. 1.4×10^{-8} N
d. 1.6×10^{-8} N
e. 2.5×10^{-8} N